INJURIES — TY

Forensic Medicine | 3rd Year MBBS

Hi Sammy!
Injuries chapter =
THE most asked topic!
Tables are key here!

Injuries | Weapons | Forensic Medicine | 3rd Year

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Mechanical Injuries — Basics

Abrasion vs Contusion vs Laceration = classic comparison question!

INJURY (Sec 44 IPC)

"Any harm whatever illegally caused to any person, in body, mind, reputation, or property."

Abrasion vs Contusion vs Laceration

Feature	Abrasion	Contusion	Laceration
Definition	Skin surface scraped	Bleeding under intact skin	Torn/split skin by blunt
Bleeding	Minimal/oozing	Under skin (no ext. bleed)	Present, moderate
Edges	Irregular, raw	No breach of skin	Irregular, ragged, bruised
Hair follicles	Damaged	Intact	Intact (crushed, not cut)
Healing	No scar (if superficial)	Color changes (timeline!)	Scar formation
Weapon	Rough surface, friction	Blunt force	Heavy blunt force
Age estimate	Scab formation (24-48hr)	Color changes (Reddy)	Not reliable

Contusion (Bruise) — Color Changes with Time

Time Period	Color	Pigment
Fresh (0-2 days)	Red/Blue	Deoxygenated hemoglobin
2-4 days	Blue/Bluish-black	Hemoglobin breakdown begins
5-7 days	Bluish-green	Biliverdin formation
7-12 days	Greenish-yellow	Biliverdin + bilirubin
12-14 days	Yellow/Brown	Hemosiderin (iron)
2-4 weeks	Normal (disappears)	Complete absorption

Types of Abrasion

Type	Description
Scratch/Linear	Caused by sharp pointed object (nail, thorn)
Graze/Sliding	Friction against rough surface (road rash, fall)
Pressure/Crushing	Compression of skin (teeth marks, ligature mark)
Impact	Force perpendicular to skin (steering wheel in RTA)

Incised, Stab + Firearm Injuries

Entry vs Exit wound
table = 100% asked!
Learn this cold!

Incised Wound vs Lacerated Wound

Feature	Incised Wound	Lacerated Wound
Weapon	Sharp cutting (knife, blade)	Blunt force (hammer, stone)
Edges	Clean, well-defined	Irregular, ragged, bruised
Hair/Vessels	Cleanly cut	Crushed, not cut
Bleeding	Profuse	Less (vessels crushed)
Healing	Primary intention, fine scar	Secondary intention, ugly scar
Depth	Uniform	Irregular, variable

Stab/Punctured wound: Depth > Length on surface. Single-edged knife = one sharp angle + one blunt angle. Double-edged = both angles sharp (dagger).

Firearm Injuries — Entry vs Exit Wound

Feature	Entry Wound	Exit Wound
Shape	Round/oval (neat)	Irregular, stellate
Size	Smaller	Larger (usually)
Edges	Inverted (inward)	Everted (outward)
Collar of abrasion	Present (grease ring)	Absent
Blackening/tattooing	May be present (close)	Absent
Clothing fibers	Carried inward	May be extruded
Bleeding	Less	More

Range of Fire — Findings on Skin

Range	Findings Around Entry Wound
Contact	Muzzle imprint, cherry-red CO color, soot in wound, stellate wound
Close (< 15cm)	Burning, blackening (soot), tattooing (unburnt powder)
Near (15-60cm)	Tattooing present, NO burning/blackening
Distant (>60cm)	Only entry wound features, no burns/tattooing/blackening

Rule of 9 + Dupuytren's classification = must know for theory exam!

Classification of Burns (Dupuytren's / Modern)

Degree	Depth	Appearance	Healing
1st	Epidermis only	Erythema, pain	Heals in 3-5 days, no scar
2nd	Epidermis+dermis	Blisters, severe pain	Heals in 2-3 weeks, minimal scar
3rd	Full thickness	White/charred, painless	Needs grafting, contractures
4th	Deeper (muscle/bone)	Charring, bone exposed	Amputation may be needed

Wallace's Rule of Nine (TBSA %)

Body Part	Adult %	Notes
Head & Neck	9%	Infant: 18% (larger head)
Each Upper Limb	9% each	Total both arms = 18%
Anterior Trunk	18%	Chest + abdomen front
Posterior Trunk	18%	Back
Each Lower Limb	18% each	Infant: 14% each (shorter legs)
Perineum / Genitalia	1%	Often missed in calculation!

Ante-mortem vs Post-mortem Burns

Feature	Ante-mortem	Post-mortem
Vital reaction	Present (redness, swelling)	Absent
Blisters	Contain albumin-rich fluid	Contain air only / absent
Soot in airways	Present (inhaled)	Absent
CO-Hb in blood	Present (>10%)	Absent or minimal
Line of redness	Present at blister base	Absent

IPC Sections Related to Injuries

Section	Offence	Punishment
319/323	Simple hurt	1 year / fine / both
320/325	Grievous hurt	7 years + fine
324	Hurt by dangerous weapon	3 years + fine
326	Grievous hurt by weapon	Life imprisonment or 10yr
304A	Death by negligence	2 years + fine
307	Attempt to murder	10 years, may extend to life

Exam Strategy + High Yield

Injuries = guaranteed
20+ marks every exam!
You've GOT this Sammy!

Most Repeated LAQs (10-12 marks)

1. Compare abrasion, contusion, and laceration (table format)
2. Describe firearm injuries — entry vs exit wound comparison
3. Classify burns. Describe rule of 9 and medicolegal importance
4. Differentiate incised wound from lacerated wound with IPC sections
5. Describe types of mechanical injuries with medicolegal importance

Most Repeated SAQs (5-6 marks)

1. Color changes in contusion with timeline
2. Entry wound vs exit wound (5 differences)
3. Types of abrasion with examples
4. Ante-mortem vs post-mortem burns
5. Stab wound — features and medicolegal importance
6. Rule of Nine with diagram

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Hair intact, edges ragged	Lacerated wound (blunt force)
Hair cleanly cut, sharp edges	Incised wound (sharp weapon)
Depth > surface length	Stab wound
Collar of abrasion present at	Entry wound (firearm)
Cherry-red discoloration in burn	CO poisoning (contact shot)
Most reliable sign of ante-mortem burn	Soot in airways / CO-Hb >10%
Pugilistic attitude in burns	Heat stiffening (NOT ante-mortem violence)
Tattooing in firearm wound	Close range (unburnt powder particles)
IPC section for grievous hurt	320 (definition) / 325 (punishment)
Rule of 9 — perineum	1% TBSA

♥ **Understanding wounds helps you heal and seek justice, Sammy!**
Your knowledge will save lives and protect the innocent!